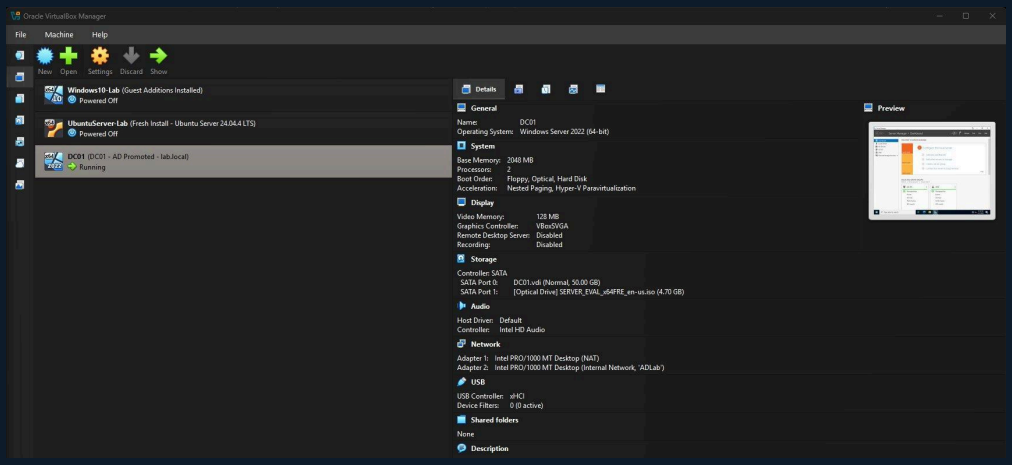
Active Directory
Home Lab

Enterprise Network Simulation from Scratch

Technical Project Documentation | Luis Moreno | 2026



VirtualBox Manager — DC01 running alongside CLIENT01

Domain	lab.local
Domain Controller	DC01 — Windows Server 2022
Client Machine	CLIENT01 — Windows 10
Platform	VirtualBox 7.2 on AMD Ryzen 7 5800XT
Network	Internal Network (ADLab) + NAT

1. Project Overview

Every morning at a Fortune 50 financial institution I'd log into my work laptop and see something like "AD: Authenticating..." flash across the screen and think, what even is that? Turns out it was Active Directory, the silent overlord of every corporate network, essentially saying "You shall not pass" to anyone without valid credentials. It controlled who I was, what I could access, and what my machine was allowed to do. So naturally, I built my own.

This lab simulates a real enterprise Windows domain network inside VirtualBox. Starting from two empty virtual machines and a downloaded ISO, I built a fully functioning Active Directory environment, a Windows Server 2022 Domain Controller hosting the domain lab.local, a Windows 10 client machine joined to that domain, user accounts managed via GUI and PowerShell, Group Policy Objects enforcing security restrictions, and audit logging capturing every login event.

No click-through tutorials. Built step by step, broken things included, with every troubleshooting moment teaching something real. This is the infrastructure that exists inside every enterprise I'll ever work in.

Live Project Assets

- Domain Controller: DC01 — Windows Server 2022 Standard Evaluation
- Client Machine: CLIENT01 — Windows 10 (domain-joined to lab.local)
- Domain: lab.local | NetBIOS: LAB
- Network: VirtualBox Internal Network "ADLab" (192.168.10.0/24)
- DC01 IP: 192.168.10.1 (static) | CLIENT01 IP: 192.168.10.10 (static)

2. Technology Stack

Technology	Purpose
Windows Server 2022 Standard Evaluation	Domain Controller OS: AD DS, DNS Server
Active Directory Domain Services (AD DS)	Core directory service: authentication, authorization, policy
DNS Server	Name resolution for lab.local: auto-installed with AD DS
Windows 10 (Build 19045)	Domain-joined client machine: end-user simulation
Group Policy Management	GPO creation, linking, and enforcement across the domain
PowerShell (AD Module)	User lifecycle: create, reset, disable, bulk operations
Event Viewer / Security Log	Audit log review: Event IDs 4624, 4625, 4634, 4740
VirtualBox 7.2	Hypervisor: hosts both VMs with snapshot support
Internal Network (ADLab)	Isolated virtual switch connecting DC01 and CLIENT01
NAT Adapter	Internet access for DC01 during setup and role installation

3. Lab Architecture

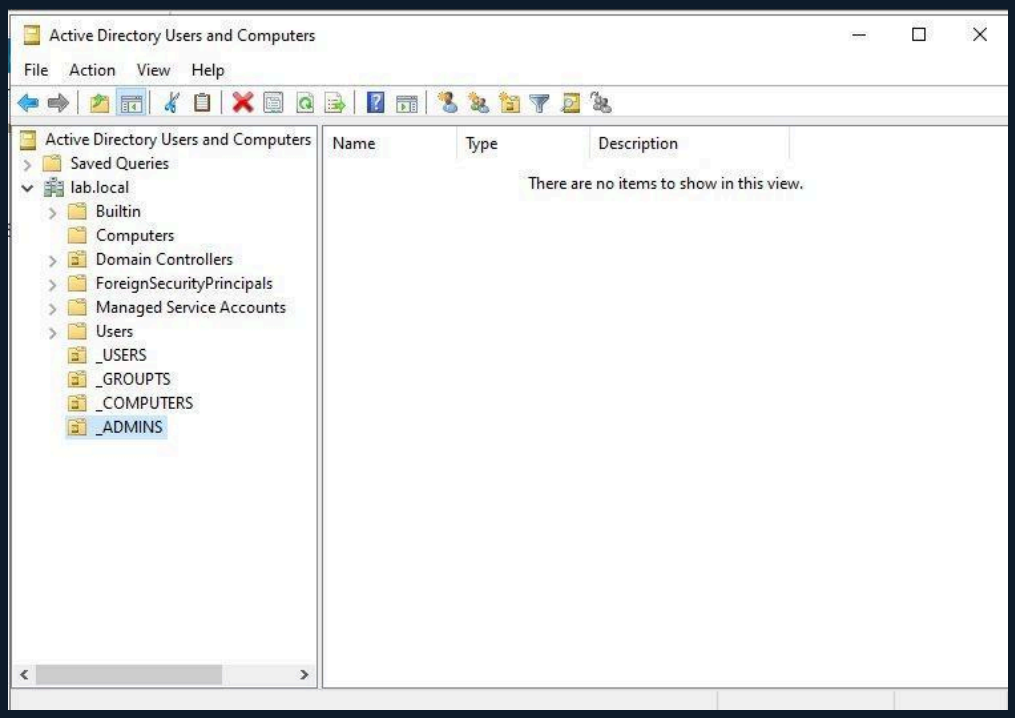
3.1 Network Topology

Both VMs share a VirtualBox Internal Network named "ADLab", an isolated virtual switch that allows the machines to communicate without touching the host network. DC01 also has a NAT adapter for internet access during setup. Once the domain was built, all AD traffic flows entirely through the internal network.

Component	DC01	CLIENT01
OS	Windows Server 2022 Standard	Windows 10 (Build 19045)
Role	Domain Controller, DNS Server	Domain Member Workstation
Adapter 1	NAT (10.0.2.15)	Internal Network ADLab
Adapter 2	Internal Network ADLab	—
Static IP	192.168.10.1	192.168.10.10
DNS	127.0.0.1 (self)	192.168.10.1 (DC01)
RAM	2048 MB	2048 MB (4GB assigned)

3.2 Active Directory Structure

The domain lab.local uses a clean OU structure modeled after real enterprise AD design. Each OU holds specific object types, allowing Group Policy to be applied surgically, standard users get locked down, admins stay unrestricted.



ADUC showing OU structure under lab.local

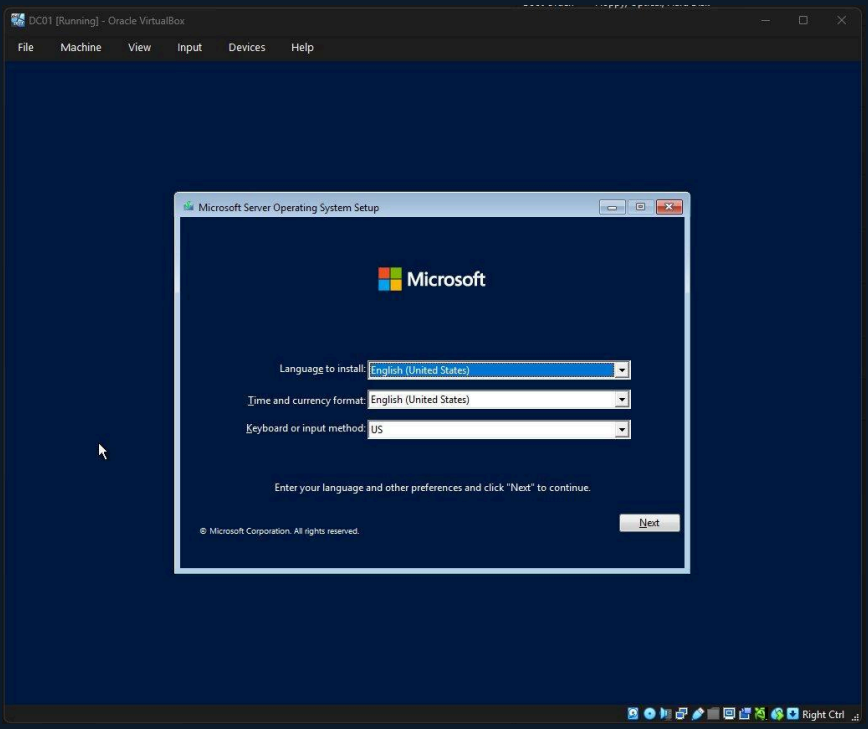
3.3 Authentication Flow

When a user logs into CLIENT01, the machine contacts DC01's DNS to locate a Domain Controller for lab.local. DC01 authenticates the credentials against the AD database (NTDS.dit), issues a Kerberos ticket, and grants access based on the user's group memberships and applicable GPOs. This is exactly what was happening every morning at a Fortune 50 financial institution.

4. Build Phases

4.1 Phase 1 — VM Creation & OS Install

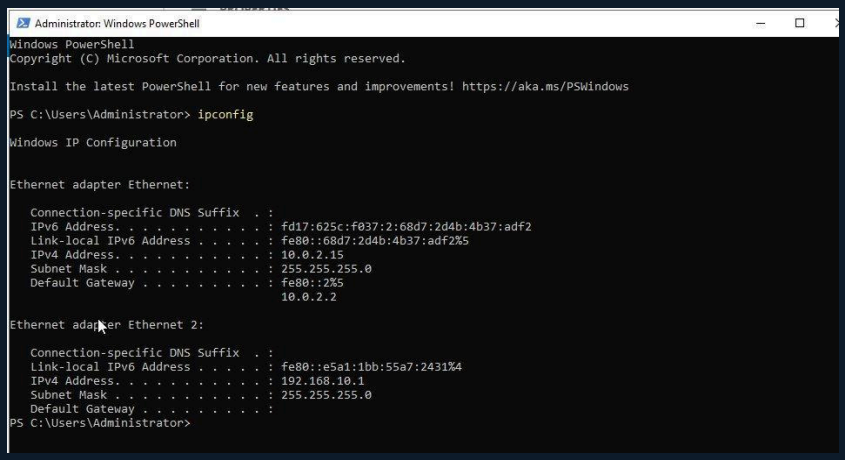
Created DC01 in VirtualBox with 2GB RAM, 2 vCPUs, and a 50GB dynamically allocated disk. Attached the Windows Server 2022 evaluation ISO and configured two network adapters: NAT for internet access and Internal Network (ADLab) for domain traffic. Installed Windows Server 2022 Standard Evaluation with Desktop Experience, the full GUI version required for sysadmin work.



Windows Server 2022 installer booting from ISO inside DC01

4.2 Phase 2 — Static IP & Server Rename

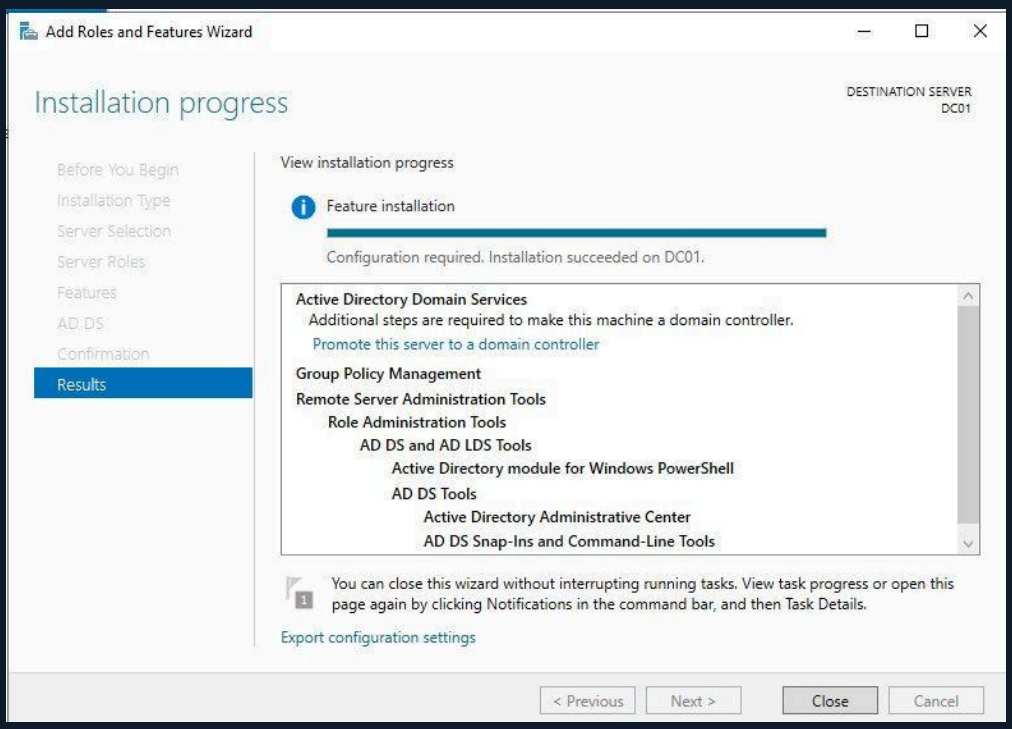
A Domain Controller must have a static IP, if the DC's IP changes, clients can't find it for authentication. Renamed the server to DC01 before promoting (name changes after promotion are painful and can break domain services). Configured Ethernet 2 with static IP 192.168.10.1 and DNS set to 127.0.0.1, the loopback address, pointing the DC to itself as its own DNS server.



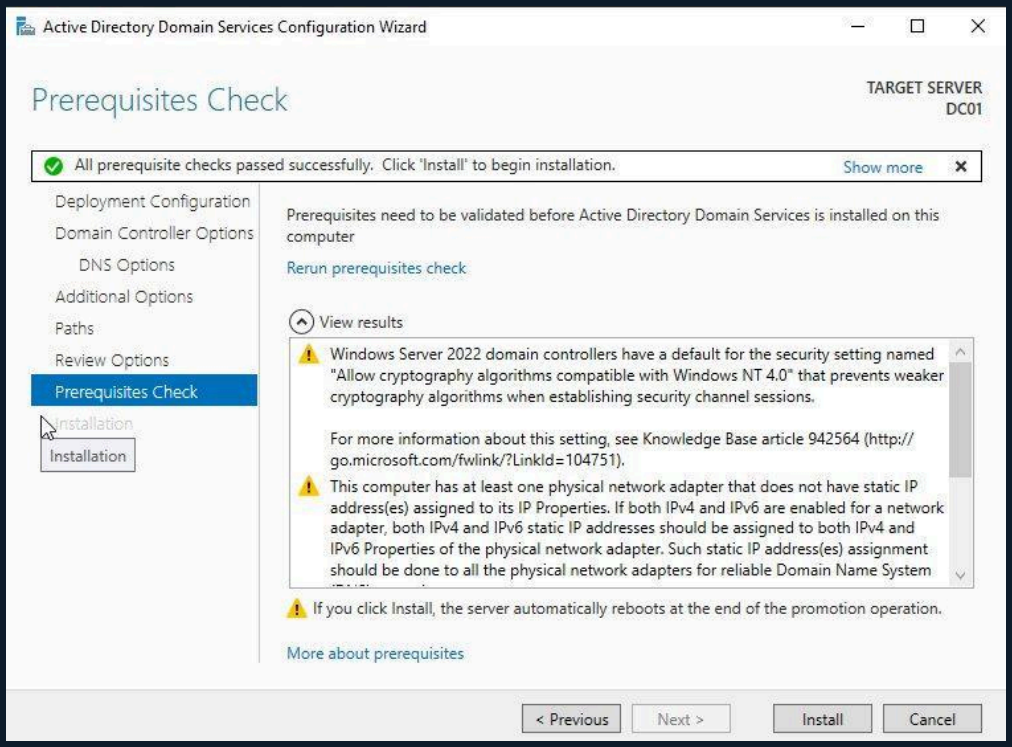
ipconfig confirming static IP 192.168.10.1 on Ethernet 2

4.3 Phase 3 — AD DS Role Install & Domain Promotion

Installed the Active Directory Domain Services role via Server Manager, along with its dependencies: Group Policy Management, Active Directory PowerShell Module, and AD DS Snap-Ins. After installation, promoted the server to Domain Controller, creating a new forest with root domain lab.local. DNS was automatically installed. Server rebooted and came back up as LAB\Administrator.



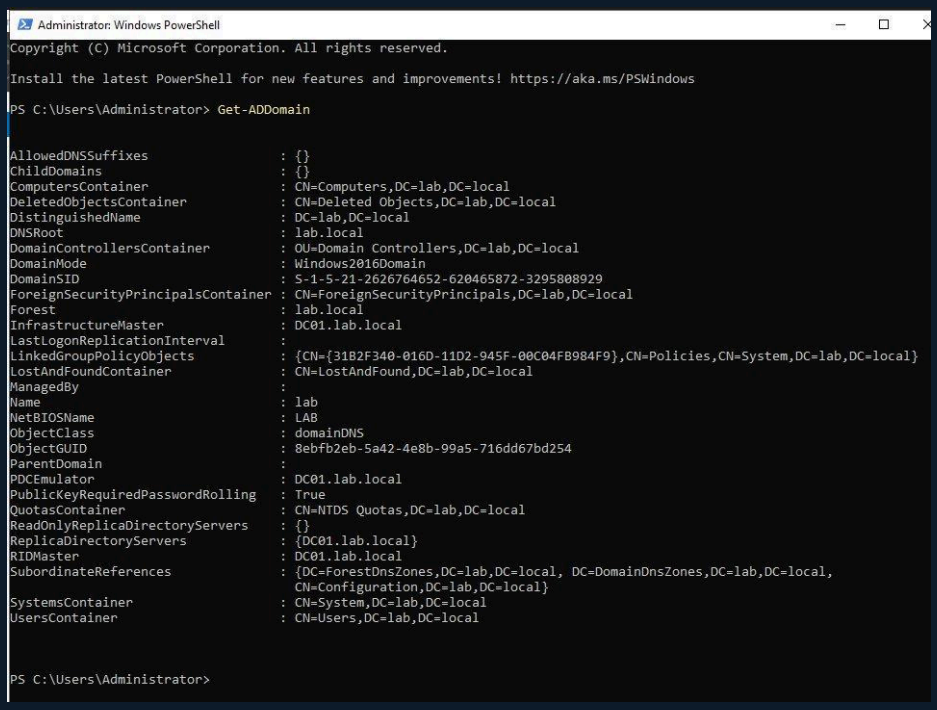
AD DS installation complete — promote link visible



Prerequisites check passed — green light for domain promotion

4.4 Phase 4 — Domain Verification

After the server rebooted as a Domain Controller, verified the domain using PowerShell. The Get-ADDomain command returned the full domain configuration including the PDC Emulator, DNS root, and NetBIOS name — confirming lab.local was fully operational.

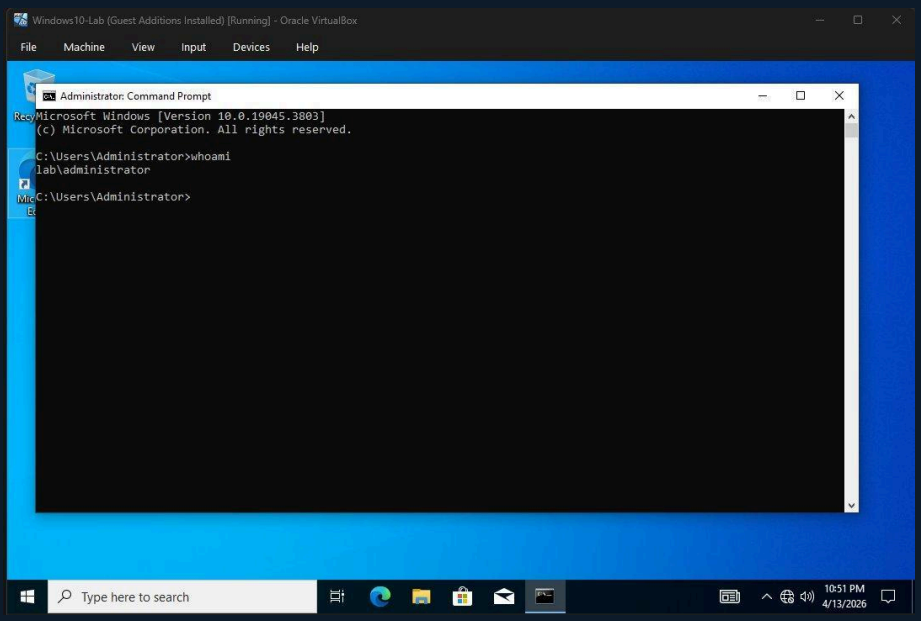


```
Administrator: Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Administrator> Get-ADDomain


AllowedDNSSuffixes                 : {}
ChildDomains                       : {}
ComputersContainer                 : CN=Computers,DC=lab,DC=local
DeletedObjectsContainer            : CN=Deleted Objects,DC=lab,DC=local
DistinguishedName                  : DC=lab,DC=local
DNSRoot                            : lab.local
DomainControllersContainer         : OU=Domain Controllers,DC=lab,DC=local
DomainMode                         : Windows2016Domain
DomainSID                          : S-1-5-21-2626764652-620465872-3295808929
ForeignSecurityPrincipalsContainer : CN=ForeignSecurityPrincipals,DC=lab,DC=local
Forest                             : lab.local
InfrastructureMaster               : DC01.lab.local
LastLogonReplicationInterval       :
LinkedGroupPolicyObjects           : {CN={31B2F340-016D-11D2-945F-00C04FB984F9},CN=Policies,CN=System,DC=lab,DC=local}
LostAndFoundContainer              : CN=LostAndFound,DC=lab,DC=local
ManagedBy                          :
Name                               : lab
NetBIOSName                        : LAB
ObjectClass                        : domainDNS
ObjectGUID                         : 8ebfb2eb-5a42-4e8b-99a5-716dd67bd254
ParentDomain                       :
PDCEmulator                        : DC01.lab.local
PublicKeyRequiredPasswordRolling   : True
QuotasContainer                    : CN=NTDS Quotas,DC=lab,DC=local
ReadOnlyReplicaDirectoryServers    : {}
ReplicaDirectoryServers            : {DC01.lab.local}
RIDMaster                          : DC01.lab.local
SubordinateReferences              : {DC=ForestDnsZones,DC=lab,DC=local, DC=DomainDnsZones,DC=lab,DC=local,
                                     CN=Configuration,DC=lab,DC=local}
SystemsContainer                   : CN=System,DC=lab,DC=local
UsersContainer                     : CN=Users,DC=lab,DC=local



PS C:\Users\Administrator>
```

Get-ADDomain output — lab.local live with DC01 as PDC Emulator

4.5 Phase 5 — CLIENT01 Domain Join

Configured the Windows 10 VM with static IP 192.168.10.10 and DNS pointing to DC01 (192.168.10.1). Verified connectivity with ping before attempting domain join. Renamed the machine CLIENT01 and joined it to lab.local using domain admin credentials. On first login as LAB\jsmith, CLIENT01 created a domain profile — exactly what happens when a new employee first logs into a corporate workstation.



```
Windows10-Lab (Guest Additions Installed) [Running] - Oracle VirtualBox
File  Machine  View  Input  Devices  Help

Administrator: Command Prompt
Microsoft Windows [Version 10.0.19045.3803]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>whoami
lab\administrator

C:\Users\Administrator>

Type here to search
10:51 PM
4/13/2026
```

whoami returning lab\administrator on CLIENT01

5. Active Directory Administration

5.1 Organizational Units

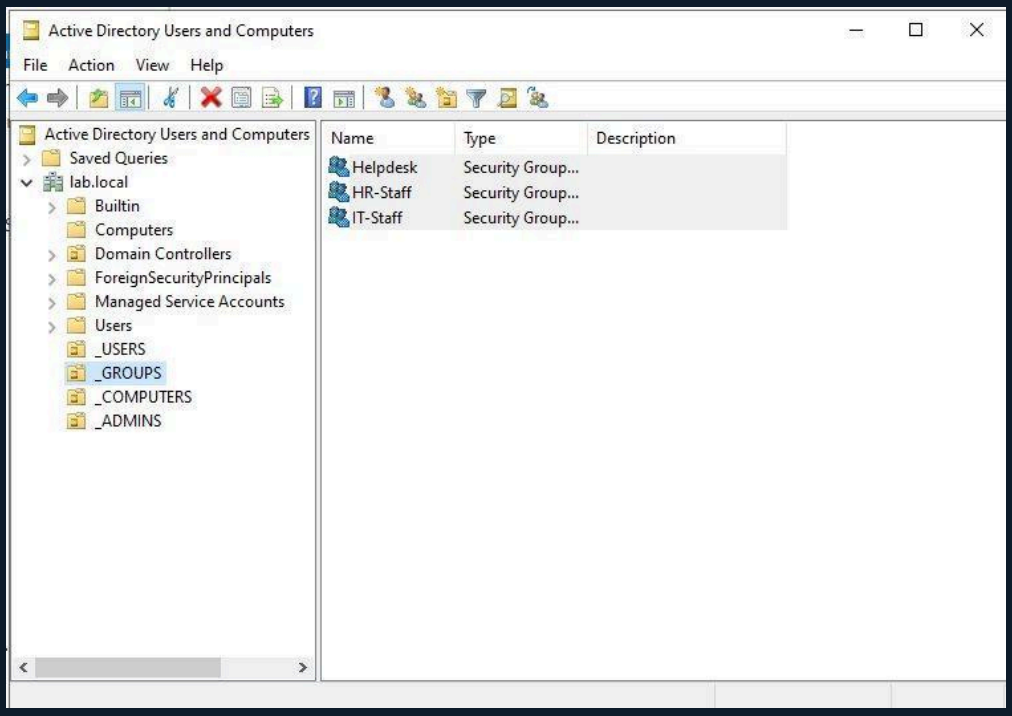
Created four OUs directly under lab.local to organize AD objects by type and role. The underscore prefix keeps them sorted at the top, a standard sysadmin convention. OUs serve as the attachment points for Group Policy and delegation.

- _USERS — standard domain users (jsmith, mlopez)
- _ADMINS — privileged accounts (itadmin)
- _GROUPS — security groups (IT-Staff, HR-Staff, Helpdesk)
- _COMPUTERS — workstations (CLIENT01)

5.2 Users & Security Groups

Created domain user accounts via both GUI (ADUC) and PowerShell. Security groups were created with Global scope and Security type, the standard for access control in enterprise environments.

User	Username	OU	Group
John Smith	jsmith	_USERS	HR-Staff
Maria Lopez	mlopez	_USERS	Helpdesk
IT Admin	itadmin	_ADMINS	IT-Staff

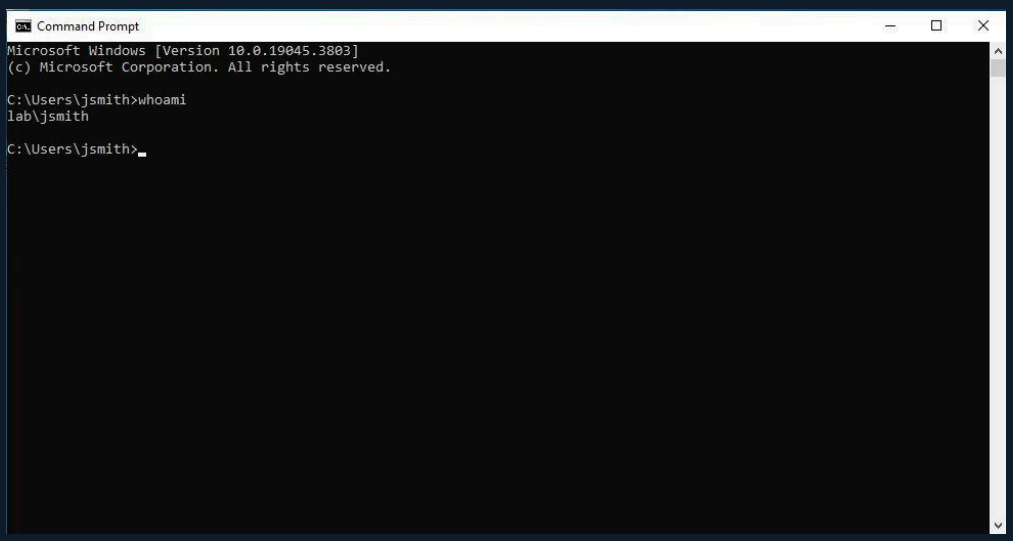


Security groups in the _GROUPS OU

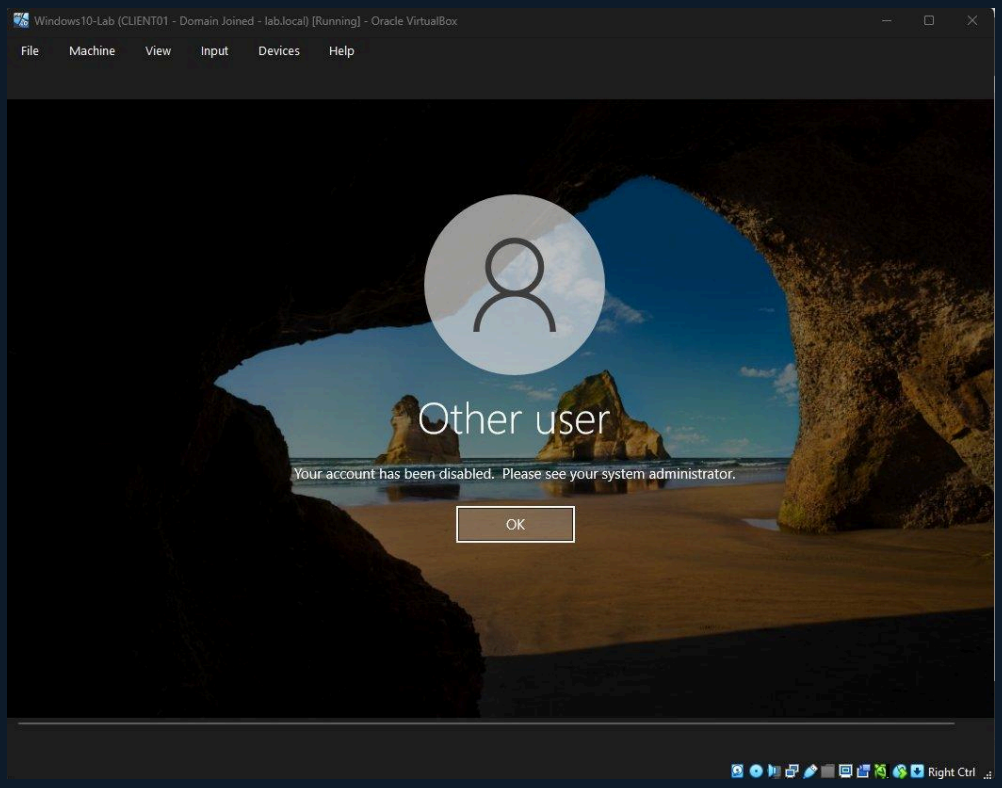
5.3 User Lifecycle Management via PowerShell

Practiced the most common helpdesk AD tasks using PowerShell:

```
# Reset password
$pass = ConvertTo-SecureString "NewP@ss123" -AsPlainText -ForceSet-ADAccountPassword -Identity
"jsmith" -Reset -NewPassword $pass

# Disable / enable account
Disable-ADAccount -Identity "jsmith"Enable-ADAccount -Identity "jsmith"

# Unlock account & check status
Unlock-ADAccount -Identity "jsmith"Get-ADUser -Identity "jsmith" -Properties LockedOut | Select
Name, LockedOut
```



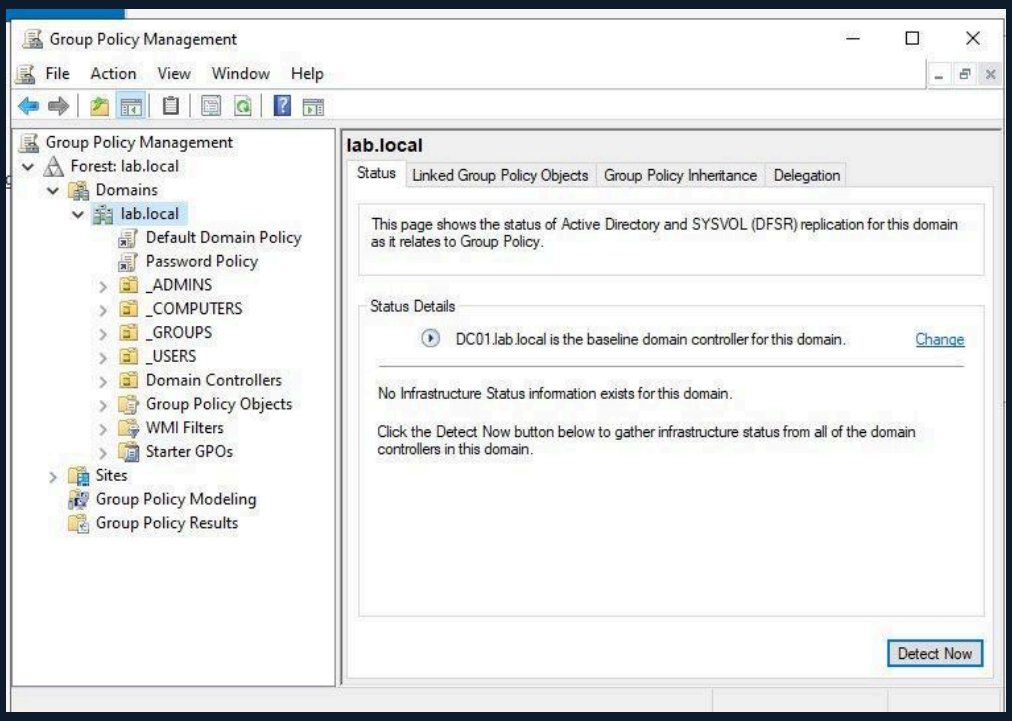
whoami returning lab\jsmith — password reset verified



Account disabled message — Disable-ADAccount working instantly

6. Group Policy Objects (GPOs)

Group Policy is how sysadmins enforce configuration and security settings across every machine in the domain, from one place, without touching individual computers. Three GPOs were created and tested.



Group Policy Management — three GPOs configured

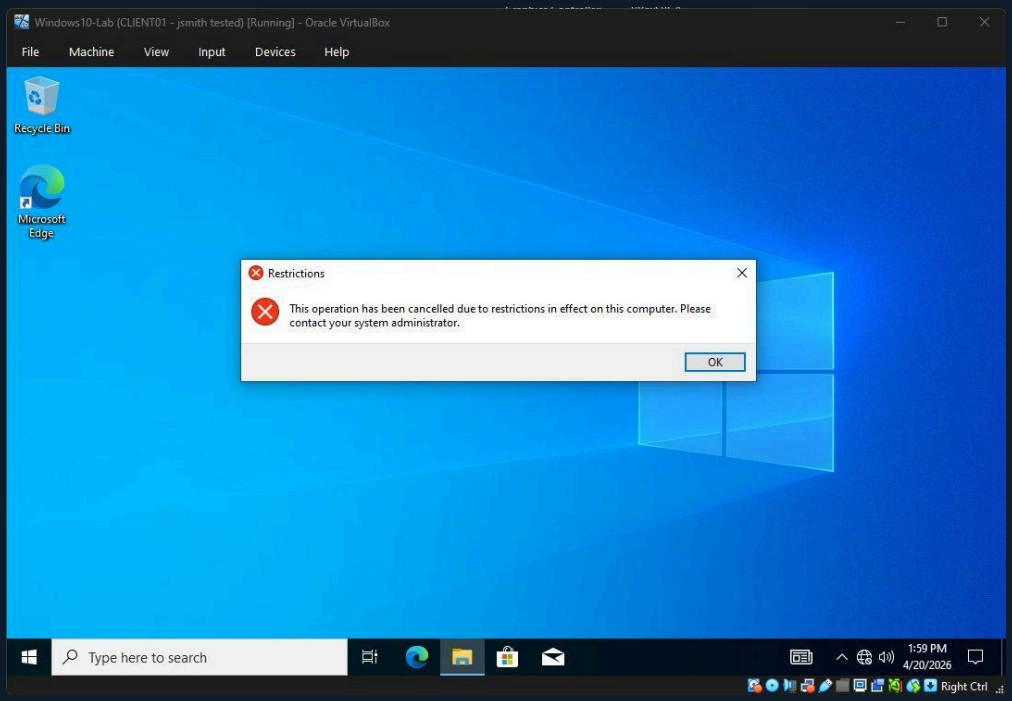
6.1 Password Policy (linked to lab.local)

Setting	Value
Minimum password length	10 characters
Password complexity	Enabled
Maximum password age	90 days
Minimum password age	30 days
Enforce password history	5 passwords remembered

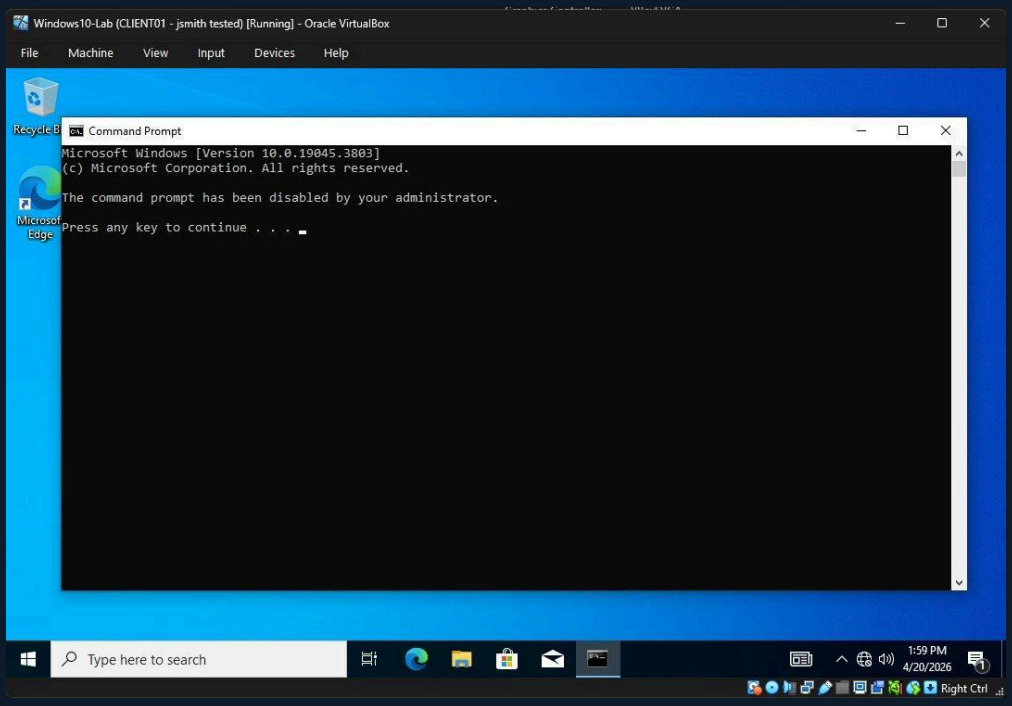
6.2 User Restrictions (linked to _USERS OU)

Applied only to the _USERS OU, blocking standard users from sensitive system tools while leaving admin accounts unaffected. Proven by logging in as jsmith and attempting to open both Control Panel and CMD.

- Prohibit access to Control Panel and PC settings — Enabled
- Prevent access to the command prompt — Enabled
- Prevent access to registry editing tools — Enabled



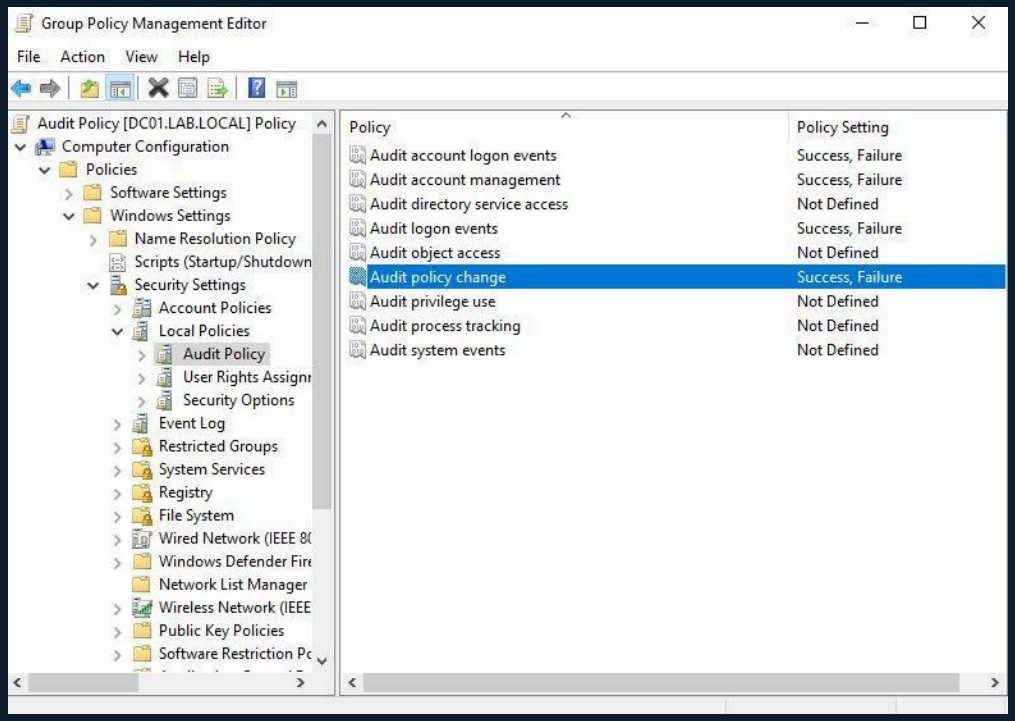
Control Panel blocked by User Restrictions GPO



CMD blocked — command prompt disabled by administrator

7. Audit Logging & Security Events

This is where the lab crosses from IT into cybersecurity territory. Configured a domain-wide Audit Policy GPO to log authentication events across every machine in lab.local, the same type of logging that Splunk ingested at a Fortune 50 financial institution.



Audit Policy GPO — four settings set to Success and Failure

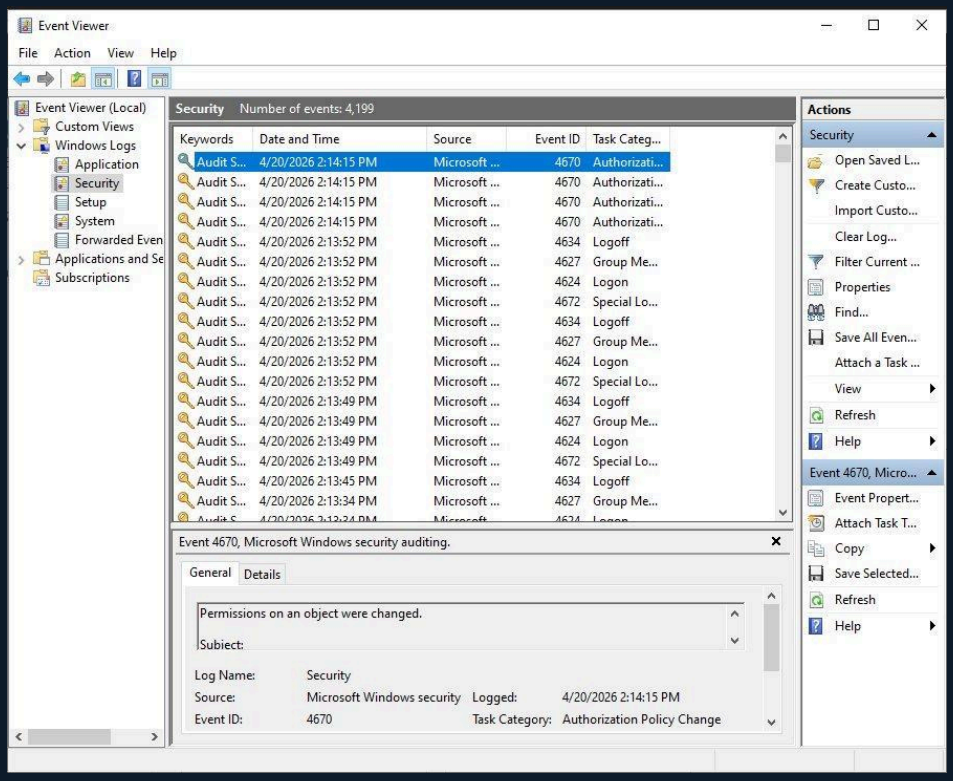
7.1 Audit Settings Configured

Audit Policy	Setting	Why It Matters
Audit account logon events	Success, Failure	Tracks all authentication attempts
Audit account management	Success, Failure	Tracks user creation, deletion, changes
Audit logon events	Success, Failure	Tracks local and network logons
Audit policy change	Success, Failure	Tracks changes to security settings

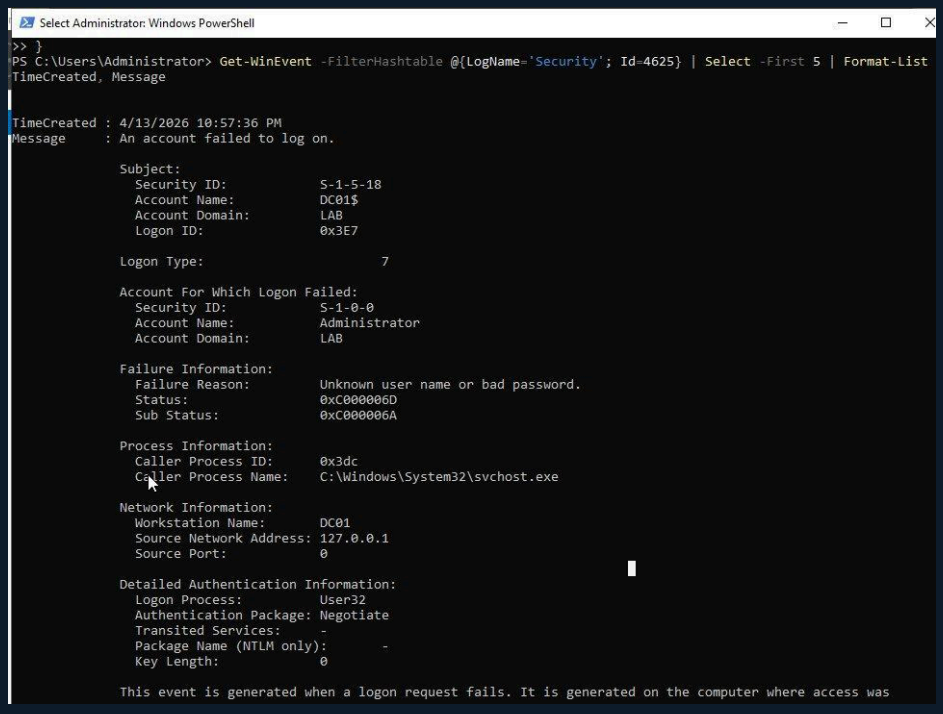
7.2 Brute Force Simulation & Event Log Review

Simulated a brute force attack by sending 5 failed authentication attempts for jsmith using a PowerShell loop. Then queried the Security Event Log for Event ID 4625 (failed logon), the same alert that SOC analysts watch for in Splunk.

```
Get-WinEvent -FilterHashtable @{LogName="Security"; Id=4625} | Select -First 5 | Format-List
TimeCreated, Message
```



Event Viewer Security Log — 4,199 events captured



Event ID 4625 — failed logon detail with source IP and account info

7.3 Critical Event IDs

Event ID	Meaning	Security Significance
4624	Successful logon	Normal auth — baseline for anomaly detection
4625	Failed logon	Multiple rapid failures = brute force attack
4634	Account logoff	Session tracking
4720	User account created	Unauthorized creation = insider threat
4740	Account locked out	Lockout trigger — often indicates attack

8. Skills Demonstrated

Skill Area	Details
IT Infrastructure	Built enterprise-grade domain network from scratch in VirtualBox
Active Directory	Domain creation, OU design, user/group management, domain join
Windows Server Admin	Server Manager, role installation, static IP, DNS configuration
PowerShell	AD module cmdlets — user creation, password reset, account management
Group Policy	GPO creation, OU-scoped linking, policy enforcement and verification
Network Configuration	Static IPs, DNS resolution, internal networking, adapter management
Security & Auditing	Audit policy design, Event Viewer analysis, brute force simulation
Troubleshooting	DNS failures, domain join errors, firewall rules, policy conflicts
Virtualization	VirtualBox VM creation, snapshots, multi-VM networking, ISO management
Identity & Access Mgmt	Least privilege, OU delegation, account lockout, access control

9. Future Lab Expansions

- Second Domain Controller (DC02) for redundancy and failover testing
- DHCP Server role — dynamic IP assignment for additional client VMs
- File Server with shared drives — NTFS permissions and access control
- Splunk Universal Forwarder — ship Windows Event Logs to a SIEM for real-time alerting
- Kali Linux VM — AD enumeration and attack simulation (BloodHound, enum4linux)
- Fine-Grained Password Policies — different requirements per security group
- Remote Desktop Services — RDP access control and session management

10. About the Developer

Luis Moreno is a Computer Engineering graduate and former Software Engineering Associate at Wells Fargo, currently pursuing an MS in Computer Science with a Cybersecurity concentration at Arizona State University (Ira A. Fulton Schools of Engineering, online, starting Summer 2026). Actively transitioning into Seeking IT, DevOps, SRE, Software Engineering Roles

GitHub	https://github.com/Luis-Morenoo
Portfolio	https://luis-morenoo.github.io
LinkedIn	https://linkedin.com/in/luis-moreno11
Email	luismorenosofteng@gmail.com
Certifications	CompTIA A+ (in progress), Network+, Security+ (planned), AWS Cloud Practitioner (Planned), Oracle Java SE (Planned)

This lab was built session by session, question by question — not from a tutorial, but from actually understanding why each step mattered. Every error was a lesson. Every working command was a win.